Mathematical Methods in the Applied Sciences

WILEY

RESEARCH ARTICLE

Love Wave Propagation in an Isotropic Thermoelastic Layer With Rigid Boundary Lying Over a Nonhomogeneous Elastic Half-Space

Abhilasha Saini | Ravinder Kumar

Department of Mathematics, University Institute of Sciences, Chandigarh University, Mohali, Punjab, India

Correspondence: Ravinder Kumar (ravinderpoonia25@gmail.com)

Received: 11 June 2025 | **Revised:** 30 July 2025 | **Accepted:** 13 August 2025

Keywords: inhomogeneity | interface irregularity | Love waves | nonhomogeneous half-space | thermoelasticity

ABSTRACT

This paper investigates the propagation of Love waves in an isotropic thermoelastic layer overlying a nonhomogeneous elastic half-space with a parabolic irregular interface. The study develops an analytical framework incorporating thermoelastic effects, material inhomogeneity, and the interface irregularity. The dynamical equations for each medium are taken using Biot's theory of elasticity, taking temperature-dependent variables for the thermoelastic layer. The dispersion equation for Love waves is developed using Fourier transformations and perturbation methods, and it shows the relation between phase velocity, wave number, and various parameters of the medium. Numerical simulations are carried out using MATLAB to analyze the influence of key parameters, including the inhomogeneity factor, irregularity depth-to-layer height ratio, and the position of the source causing irregularity. The graphical results demonstrate that the phase velocity decreases with an increase in wave number, and for the higher values of inhomogeneity parameters in half-spaces, material heterogeneity dominates and suppresses the influence of interface irregularity. These findings provide valuable insights for seismological applications, particularly in understanding wave propagation in heterogeneous, thermally affected subsurface structures.

MSC2020 Classification: 74J15, 74F05, 74B05, 74H10

1 | Introduction

Seismology, the study of seismic waves generated by natural events like earthquakes or by artificial means such as manmade explosions, plays a crucial role in exploring the internal structure and dynamic behavior of the Earth. These waves, categorized as body waves or surface waves, help reveal critical information about subsurface elastic properties, compositional heterogeneity, and active deformation mechanisms. Researchers analyze seismic waves to reconstruct geological features, quantify tectonic stress regimes, and assess seismic hazard potential. Among surface waves, Love waves have gathered considerable interest in seismology, geophysics, and engineering because of their distinctive characteristics and wide range of practical applications. Love waves play a vital role in various applications, including earthquake hazard assessment, subsurface imaging, and nondestructive material testing. Their sensitivity to variations in material properties, layer thickness, and subsurface irregularities makes them particularly useful for investigating complex geological environments. Classical studies on Love wave propagation have largely considered idealized models involving homogeneous and isotropic elastic media. However, real-world scenarios are significantly more complex, often including material inhomogeneity, porosity, thermoelastic effects, and irregular interface geometries, all of which influence wave behavior. This fundamental gap between theoretical predictions and observed wave behavior

© 2025 John Wiley & Sons Ltd.

in complex media highlights the need to improve both forward and inverse modeling approaches in seismic studies. Research on Love waves began with foundational work by Love [1], who first described shear horizontal wave propagation in a homogeneous isotropic elastic layer over a semi-infinite substrate. Subsequent studies extended this work to include multilayered systems and the effects of various parameters like anisotropy and viscoelasticity. The classical analytical solutions provided by Love have been extended by numerical techniques (such as finite element and spectral methods) to address more complex geometries and material configurations. Building on this foundation, Ewing et al. [2] further advanced the understanding of surface wave propagation in elastic media, highlighting the critical influence of layer thickness and substrate rigidity on wave velocity. The theoretical framework for Love wave propagation has been progressively developed through pivotal contributions. Graff [3] systematically derived the dispersion relations governing frequency-dependent phase velocity in elastic media. Power [4] investigated the propagation of linear waves at the free surface of saturated porous media, framing the problem as a complex eigenvalue formulation rather than a classical Sturm–Liouville problem. He emphasized the importance of properly characterizing the completeness of eigenfunctions, particularly in parameter regimes where traditional assumptions may fail. Achenbach [5] contributed to the field by developing a rigorous elasto-dynamic formulation for wave propagation in stratified systems, quantifying how layer interfaces modify surface wave characteristics. Chattopadhyay and Kar [6] investigated the propagation of Love-type waves by solving the boundary value problem for point-source excitation in an isotropic elastic material under initial stress. While these foundational works established critical theoretical benchmarks, the idealized assumptions of homogeneity and isotropy cannot adequately address the complex wave phenomena observed in realistic geological settings.

Among the various factors contributing to the complexity of geological media, thermoelastic coupling between thermal and mechanical fields introduces critical modifications to wave propagation, particularly in media where temperature variations influence mechanical responses. In heterogeneous media, this interaction reveals two primary mechanisms: temperature-dependent stiffness variations that modify phase velocities and thermal diffusion-induced attenuation that disproportionately affects high-frequency components. Additionally, geometric irregularities cause wave scattering and diffraction that disrupt coherent propagation. These topographic features redistribute seismic energy through amplitude focusing/defocusing effects, generation of secondary wavefronts, and frequency-dependent path modulation, all of which compound with thermoelastic and heterogeneity induced distortions to produce observed wavefield complexities. Interface irregularities can cause wave scattering, diffraction, and mode conversion, producing complex dispersion characteristics. These features are common in natural settings like fault zones and sedimentary basins. Studying their effects is essential for realistic modeling wave behavior in such environments. Biot [7] developed a foundational framework for thermoelasticity, emphasizing the coupling between thermal and mechanical fields using the theory of irreversible thermodynamics. Biot's theory introduced the concept of generalized free energy, serving as a thermoelastic potential, and established a variational approach to describe energy dissipation in deformable media. This work laid the theoretical groundwork for contemporary models of coupled thermal mechanical behavior in solid materials. To address the nonphysical prediction of infinite thermal signal speed inherent in classical heat conduction models, Lord and Shulman [8] proposed a generalized theory incorporating a modified heat conduction equation that ensures finite propagation speed of thermal signals. Green and Lindsay [9] advanced the theory by introducing two thermal relaxation times, improving its accuracy, while Green and Naghdi [10] later developed a more generalized model that allows undamped thermal wave propagation, making it particularly suitable for high-frequency thermal processes. Building on these developments, Bear et al. [11] formulated a comprehensive set of macroscopic equations for saturated thermoelastic porous media under nonisothermal conditions, demonstrating that the fluid and solid displacements are governed by two coupled wave equations induced by abrupt thermal and pressure disturbances. Kumar and Saini [12, 13] investigated the influence of key material parameters, such as anisotropy, initial stress, and inhomogeneity, on the propagation of Love waves in layered media featuring parabolic interfacial irregularities. Kumar et al. [14] investigated Love-type wave propagation in a thermoelastic medium with double porosity structures and derived a modified dispersion relation accounting for thermal and porosity effects. Their results revealed the existence of two distinct wave fronts, with the second front showing higher phase velocity influenced by thermal properties, while porosities led to a reduction in wave speed. Many researchers have investigated the effects of geometric and interfacial irregularities on surface wave propagation, demonstrating how such features, when coupled with material complexity, can lead to significant alterations in wave behavior. Such as Kumar et al. [15, 16] investigated the impact of different boundary conditions and a rectangular interface irregularity on the propagation of surface waves in multilayered media. Saini and Kumar [17, 18] analyzed the effect of parabolic interfacial irregularity on the propagation of Love waves in a transversely isotropic fluid-saturated porous layer. Recent studies have also addressed wave scattering due to interface undulations in complex media. Lalvohbika and Singh [19] analyzed qSH-waves across a corrugated interface between dissimilar nematic elastomers, highlighting the dependence of phase velocity and energy ratios on material properties and interface geometry. Singh [20] studied Love wave propagation in an isotropic layer bounded by corrugated surfaces, deriving the dispersion relation and showing how boundary irregularities influence wave behavior. More recently, Singh and Oinam [21] applied the Haskell matrix method to multilayered magneto-thermoelastic media, demonstrating the emergence of two distinct Love-type wavefronts arising from electromagnetic and thermal effects. Several other researchers have contributed to this field, including Kumar et al. [22], Sharma et al. [23], Kumar et al. [24], Gajroiya and Sikka [25], Kumar et al. [26], Gajroiya et al. [27], Ailawalia et al. [28], and Sahrawat and Kumar [29].

The primary objective of this study is to analyze the propagation of Love waves in a thermoelastic layer overlying a nonhomogeneous elastic half-space with a parabolic interfacial irregularity generated by a point source. The model comprehensively accounts for the combined effects of thermoelasticity, inhomogeneity in the half-space, the geometry of the interfacial irregularity, and the location of the point source. To derive the dispersion

relation for Love waves, Fourier and inverse Fourier transforms are employed alongside the perturbation method. This dispersion relation is then examined to evaluate the influence of thermoelastic effects, material inhomogeneity, and interface irregularities on phase velocity and wave attenuation. Numerical simulations, supported by MATLAB-based graphical analysis, illustrate the variation of phase velocity with wave number, and surface plots are provided to depict its dependence on both wave number and point-source distance, thereby offering deeper insight into the complex wave behavior in such layered media.

2 | Problem Formulation

To formulate mathematical model for the considered problem, we assume the rectangular coordinate system (x, y, z) with x-axis parallel to the direction of Love wave propagation and z-axis oriented vertically downward. The model consists of a homogenous isothermal layer of thickness H, defined as medium $(M_1 : -H \leq z \leq 0)$, resting over a nonhomogeneous elastic half-space, defined as medium $(M_2 : 0 \leq z < \infty)$. The origin "o" of the coordinate system is located at the midpoint of the irregularity that exists at the interface of layer and half-space. The disturbance causing source is positioned on the positive z-axis at a distance of $d(d > l)$ from the origin. This configuration is illustrated in Figure 1.

The equation for the irregularity considered using Saini and Kumar [17] is

$$z = \varepsilon f(x),$$

where

$$f(x) = \begin{cases} 0, & \text{for } |x| \geq s \\ \frac{2}{s}\left(s^2 - x^2\right) & \text{for } |x| < s \end{cases}, \tag{1}$$

where $\varepsilon = \frac{l}{2s} << 1, s \neq 0$.

3 | Governing Equations

The basic equations and constitutive relations for M_1 and M_2 are given as follow:

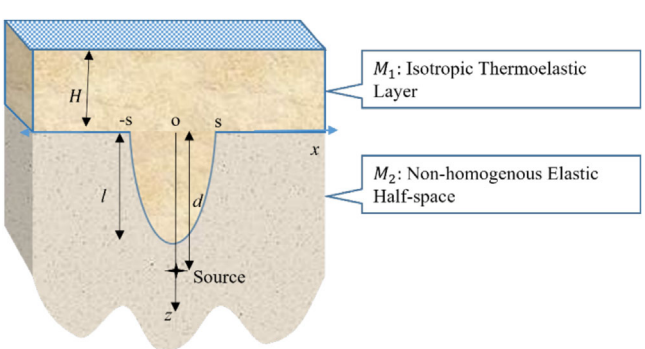

FIGURE 1 | Geometry of the problem. [Colour figure can be viewed at wileyonlinelibrary.com]

3.1 | Medium M_1

Following Lord and Shulman [8], the dynamical equations of motion for M_1 are given as follows:

$$\mu\nabla^2 u + \left(\lambda_1 + \mu_1\right)\nabla(\nabla \cdot u) - \beta\nabla\theta = \rho_1 \ddot{u}_i, \tag{2}$$

$$K\nabla^2 T_i = \beta T_0 \nabla \cdot \dot{u}_i + aT_0 \dot{T}_i, \tag{3}$$

where K is thermal conductivity, a is thermal capacity, and $\beta = 3\lambda + 2\mu$.

The constitutive relations for M_1 given by Lord and Shulman [8] are

$$\sigma_{ij} = \lambda e_{kk}\delta_{ij} + 2\mu_1 e_{ij} - \beta T_i \delta_{ij}, \tag{4}$$

where $e_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right)$.

As we have considered a two-dimensional problem and Love waves are propagating in the x-axis direction such that the basic components are independent of the y-axis, so the expressions for the displacement and temperature components are given as

$$\left.\begin{aligned} u_1 = u_3 = 0 \text{ and } u_2 = u_2(x, z, t) \\ T_1 = T_3 = 0 \text{ and } T_2 = T_2(x, z, t) \end{aligned}\right\}. \tag{5}$$

By using Equations (4) and (5) in Equations (2) and (3), we get

$$\rho_1 \frac{\partial^2 u_2}{\partial t^2} = \mu_1 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right) u_2, \tag{6}$$

$$K\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right) T_2 - c_v \frac{\partial T_2}{\partial t} = 0, , \tag{7}$$

where $c_v = aT_0$.

3.2 | Medium M_2

The dynamical equation of motion for M_2 in the absence of the body forces are taken from Ewing et al. [2]:

$$\tau_{ij,j} = \rho_2 \ddot{v}_i. \tag{8}$$

Following Saini and Kumar [17], the constitutive relation is taken as

$$\tau_{ij} = \lambda p_{kk}\delta_{ij} + 2\mu_2 p_{ij}, \tag{9}$$

where p_{ij} is defined as

$$p_{ij} = \frac{1}{2}\left(v_{i,j} + v_{j,i}\right), p_{kk} = v_{kk}. \tag{10}$$

Using $v_1 = v_3 = 0, v_2 = v_2(x, z, t)$ and Equation (9) in Equation (8), we get

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right) v_2 = \frac{1}{c_2^2}\frac{\partial^2 v_2}{\partial t^2}, \tag{11}$$

where

$$c_2^2 = \frac{\mu_2}{\rho_2}, \mu_2 = \mu^0 \exp(qz), \ \ \rho_2 = \rho_0 \exp(qz) \tag{12}$$

are inhomogeneity for the half-space for some constant values μ_2^0, ρ_2^0, and q.

4 | Boundary Conditions

The equations of motion are applied to a bounded medium, so specific conditions are introduced to describe the behavior of stresses, displacement, and thermal conditions at boundaries. These conditions are.

1. At the rigid boundary $z = -H$, the displacement vector vanishes:
$$u_2(x, z, t) = 0. \tag{13}$$
The thermal condition at the rigid boundary is 0, as we are considering the layer to be thermally insulated, that is,
$$T_2(x, z = 0, t) = 0. \tag{14}$$
2. The interface of medium M_1 and half-space M_2 is considered to be continuous that leads the displacement components and stress components to be continuous at $z = \varepsilon f(x)$, that is,
   - Continuity of displacement components are
$$u_2(x, z, t) = v_2(x, z, t). \tag{15}$$
   - Continuity of stress components is
$$\mu_1\left(\frac{\partial u_2}{\partial z} - \varepsilon f'(x)\frac{\partial u_2}{\partial x}\right) = \mu_2\left(\frac{\partial v_2}{\partial z} - \varepsilon f'(x)\frac{\partial v_2}{\partial x}\right), \tag{16}$$
   where $f'(x) = \frac{df(x)}{dx}$.

Thermal condition at the interface, $z = \varepsilon f(x)$, is again considered to be 0, as the layer is insulated, that is,
$$T_2(x, z = \varepsilon h(x), t) = 0. \tag{17}$$

5 | Analytical Solution of the Problem

Because the waves are propagating harmonically with respect to time, the harmonic form of basic variables can be taken as
$$\{u_2, v_2, T_2\}(x, z, t) = \{u_2^0, v_2^0, T_2^0\}(x, z)e^{(-i\omega t)}. \tag{18}$$

Using Equation (18), the equation of motion of each medium (2), (3), and (8) changes to
$$\left\{\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right) + \frac{\omega^2}{c_t^2}\right\}u_2^0 = 0, \tag{19}$$
$$\left\{\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right) + i\omega c_v\right\}T_2^0 = 0, \tag{20}$$
$$\left\{\frac{\partial^2}{\partial z^2} + q\frac{\partial^2}{\partial z^2} - k^2\left(1 - \frac{c_L^2}{\beta_2^2}\right)\right\}v_2^0 = 0, \tag{21}$$

where $\omega = c_L k$, c_L is the speed of the wave, k is wavenumber, and $c_L = \sqrt{\mu_1/\rho_1}$ is the speed of classical distortional wave.

The Fourier transformation for $u_2^0(x, z)$ is defined as
$$\overline{u_2^0}(\eta, z) = \int_{-\infty}^{\infty} u_2^0(x, z)e^{i\eta x}dx, \tag{22}$$
and the corresponding inverse Fourier transformation is defined as
$$u_2^0(z, x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\overline{u_2^0}(z, \eta)e^{-i\eta x}d\eta. \tag{23}$$

By using Equation (22) in Equations (19–21), a system of differential equations is obtained:
$$\frac{d^2\overline{u_2}}{dz^2} + \zeta_1^2\overline{u_2} = 0, \tag{24}$$
$$\frac{d^2\overline{T_2}}{dz^2} + \zeta_2^2\overline{T_2} = 0, \tag{25}$$
$$\frac{d^2\overline{v_2}}{dz^2} - \frac{d\overline{v_2}}{dz} - \xi^2\overline{v_2} = 0, \tag{26}$$

where ζ_1^2, ζ_2^2, and ζ_3^2 are defined as
$$\zeta_1^2 = \left(\frac{\omega^2}{c_L^2} - \eta^2\right), \zeta_2^2 = -\left(\eta^2 + \frac{i\omega c_v}{K}\right), \text{and } \xi^2 = \eta^2 - \frac{\omega^2}{\beta^2}.$$

The general solution for the differential equations (24–26) is given by
$$\left.\begin{aligned}\overline{u_2^0} &= C_1\cos(\zeta_1 z) + C_2\sin(\zeta_1 z)\\ \overline{T_2^0} &= C_3\cos(\zeta_2 z) + C_4\sin(\zeta_2 z)\end{aligned}\right\}, \tag{27}$$
$$\overline{v_2^0} = C_5\exp(-\zeta_3 z), \tag{28}$$

where $C_1, C_2, \ldots, C_5$ are arbitrary functions of η and $\zeta_3^2 = \frac{1}{2}\left(q + \sqrt{q^2 + 4\xi^2}\right)$.

Now, using the inverse Fourier transformation from Equation (23), Equations (26–28) change to
$$u_2^0(x, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(C_1\cos(\zeta_1 z) + C_2\sin(\zeta_1 z)\right)e^{-i\eta x}d\eta, \tag{29}$$
$$T_2^0(x, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(C_3\cos(\zeta_2 z) + C_4\sin(\zeta_2 z)\right)e^{-i\eta x}d\eta, \tag{30}$$
$$v_2^0(x, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(C_5 e^{-\zeta_3 z} + \frac{2}{\xi_3}e^{-\zeta_3 z}e^{\zeta_3 d}\right)e^{-i\eta x}d\eta. \tag{31}$$

The second term in the integrand in Equation (31) represents the presence of the source in the half-space at a distance of d from the origin on the z-axis.

6 | Frequency Equations

By applying the perturbation method given by Eringen and Samuels [30], $C_1, C_2, \ldots, C_5$ can be set for small positive constant ε as

$$\begin{aligned} &C_1 \cong C_1^0 + C_1^1\varepsilon, C_2 \cong C_2^0 + C_2^1\varepsilon, C_3 \cong C_3^0 + C_3^1\varepsilon, \\ &C_4 \cong C_4^0 + C_4^1\varepsilon, C_5 \cong C_5^0 + C_5^1\varepsilon, \end{aligned} \tag{32}$$

$$e^{\pm\alpha\varepsilon f} \cong 1 \pm \alpha\varepsilon f, \cos\left(\zeta_1\varepsilon f\right) \cong 1, \sin\left(\zeta_1\varepsilon f\right) \cong \zeta_1\varepsilon f. \tag{33}$$

Using the boundary condition (13) and (14), for $z = -H$ and Equation (32) in (29) and (30), we have

$$\begin{aligned} &\left(C_1^0 + C_1^1\varepsilon\right)\cos\left(\zeta_1 H\right) - \left(C_2^0 + C_2^1\varepsilon\right)\sin\left(\zeta_1 H\right) = 0, \\ &\left(C_3^0 + C_3^1\varepsilon\right)\cos\left(\zeta_2 H\right) - \left(C_4^0 + C_5^1\varepsilon\right)\sin\left(\zeta_2 H\right) = T_0. \end{aligned} \tag{34}$$

Now, applying the boundary conditions at $z = \varepsilon f(x)$ from Equations (15–17) and using Equation (32) in Equations (29–31), we obtain

$$\left(C_3^0 + C_3^1\varepsilon\right)\cos\left(\zeta_2\varepsilon f(x)\right) + \left(C_4^0 + C_4^1\varepsilon\right)\sin\left(\zeta_2\varepsilon f(x)\right) = T_0, \tag{35}$$

$$\begin{aligned} &\int_{-\infty}^{\infty}\left(\left(C_1^0 + C_1^1\varepsilon\right)\cos\left(\zeta_1\varepsilon f(x)\right) + \left(C_2^0 + C_2^1\varepsilon\right)\sin\left(\zeta_1\varepsilon f(x)\right)\right)e^{-i\eta x}d\eta \\ &= \int_{-\infty}^{\infty}\left(\left(C_5^0 + C_5^1\varepsilon\right)e^{-\zeta_3 z} + \frac{2}{\xi_3}e^{-\zeta_3 z}e^{\zeta_3 d}\right)e^{-i\eta x}d\eta, \end{aligned} \tag{36}$$

$$\begin{aligned} &\int_{-\infty}^{\infty}\begin{pmatrix} -\left(C_1^0 + C_1^1\varepsilon\right)\sin\left(\zeta_1\varepsilon f(x)\right) \\ +\left(C_2^0 + C_2^1\varepsilon\right)\cos\left(\zeta_1\varepsilon f(x)\right) \\ -\varepsilon f'(x)\left(\left(C_1^0 + C_1^1\varepsilon\right)\cos\left(\zeta_1\varepsilon f(x)\right)\right. \\ \left.+\left(C_2^0 + C_2^1\varepsilon\right)\sin\left(\zeta_1\varepsilon f(x)\right)\right) \end{pmatrix}e^{-i\eta x}d\eta \\ &= \int_{-\infty}^{\infty}\left(-\zeta_3\left(C_5^0 + C_5^1\varepsilon\right)e^{-\zeta_3\varepsilon f(x)} - 2e^{-\zeta_3\varepsilon f(x)}e^{\zeta_3 d}\right. \\ &\left. - \eta\left(C_5^0 + C_5^1\varepsilon\right)e^{-\zeta_3\varepsilon f(x)} - \frac{2}{\xi_3}\eta e^{-\zeta_3\varepsilon f(x)}e^{\zeta_3 d}\right)e^{-i\eta x}d\eta. \end{aligned} \tag{37}$$

Fourier transformation and inverse Fourier transformation for $f(x)$ is defined as

$$\overline{f}(\lambda) = \int_{-\infty}^{\infty} f(x)e^{i\lambda x}d\lambda, \tag{38}$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\overline{f}(\lambda)e^{-i\lambda x}d\lambda. \tag{39}$$

Therefore,

$$f'(x) = \frac{-i}{2\pi}\int_{-\infty}^{\infty}\lambda\overline{f}(\lambda)e^{-i\lambda x}d\lambda. \tag{40}$$

Equations (35–37) can be further simplified by using Equations (39) and (40) and perturbation approximation in (33), to

$$\left(C_3^0 + C_3^1\varepsilon\right) + \left(C_4^0 + C_4^1\varepsilon\right)\zeta_2\varepsilon f = T_0, \tag{41}$$

$$\begin{aligned} &\int_{-\infty}^{\infty}\left(C_1^0 + C_1^1\varepsilon + \left(C_2^0 + C_2^1\varepsilon\right)\zeta_1\varepsilon f\right)e^{-i\eta x}d\eta \\ &= \int_{-\infty}^{\infty}\left(\left(C_5^0 + C_5^1\varepsilon\right)\left(1 - \zeta_3\varepsilon f\right) + \frac{2}{\xi_3}\left(1 - \zeta_3\varepsilon f\right)e^{\zeta_3 d}\right)e^{-i\eta x}d\eta, \\ &\left(C_1^0 - C_5^0\right) + \varepsilon\left(C_1^1 - C_5^1\right) - \frac{2}{\xi_3}e^{\zeta_3 d} \\ &= -\left(C_5^0 + C_5^1\varepsilon\right)\zeta_3\varepsilon f(x) - \\ &\left(C_2^0 + C_2^1\varepsilon\right)\zeta_1\varepsilon f(x) - 2\varepsilon f(x)e^{\zeta_3 d}, \\ &\left(C_1^0 - C_5^0 - \frac{2}{\xi_3}e^{\zeta_3 d}\right) + \varepsilon\left(C_1^1 - C_5^1\right) = \varepsilon R_1, \end{aligned} \tag{42}$$

$$\begin{aligned} &\mu_1\left[-\left(C_1^0 + C_1^1\varepsilon\right)\zeta_1\varepsilon f + \left(C_2^0 + C_2^1\varepsilon\right)\right. \\ &\quad\left.-\varepsilon f'(x)\left(\left(C_1^0 + C_1^1\varepsilon\right) + \left(C_2^0 + C_2^1\varepsilon\right)\zeta_1\varepsilon f\right)\right] \\ &=\mu_2\left[-\zeta_3\left(C_5^0 + C_5^1\varepsilon\right)\left(1 - \zeta_3\varepsilon f\right)\right. \\ &\quad - 2\left(1 - \zeta_3\varepsilon f\right)e^{\zeta_3 d} - \eta\left(C_5^0 + C_5^1\varepsilon\right) \\ &\quad\left.\left(1 - \zeta_3\varepsilon f\right) - \frac{2}{\xi_3}\eta\left(1 - \zeta_3\varepsilon f\right)e^{\zeta_3 d}\right] \\ &\mu_1\left(C_2^0 + C_2^1\varepsilon\right) + \mu_2\zeta_3\left(C_5^0 + C_5^1\varepsilon\right) + 2\mu_2 e^{\zeta_3 d} \\ &\quad + \eta\mu_2\left(C_5^0 + C_5^1\varepsilon\right) + \frac{2}{\xi_3}\mu_2\eta e^{\zeta_3 d} \\ &=\left(C_1^0 + C_1^1\varepsilon\right)\mu_1\zeta_1\varepsilon f + \varepsilon\mu_1 f'(x) \\ &\quad\left(\left(C_1^0 + C_1^1\varepsilon\right) + \left(C_2^0 + C_2^1\varepsilon\right)\zeta_1\varepsilon f\right) \\ &\quad + \mu_2\zeta_3^2\left(C_5^0 + C_5^1\varepsilon\right)\varepsilon f + 2\mu_2\zeta_3\varepsilon f e^{\zeta_3 d} \\ &\quad + \eta\mu_2\zeta_3\varepsilon\left(C_5^0 + C_5^1\varepsilon\right)f + 2\varepsilon\zeta_3 e^{\zeta_3 d}f, \\ &\left(\mu_1 C_2^0 + \mu_2\zeta_3 C_5^0 + \eta\mu_2 C_5^0 + 2\mu_2 e^{\zeta_3 d} + \frac{2}{\xi_3}\mu_2\eta e^{\zeta_3 d}\right) \\ &\quad + \varepsilon\left(\mu_1 C_2^1 + \mu_2\zeta_3 C_5^1 + \eta\mu_2 C_5^1\right) = \varepsilon R_2, \end{aligned} \tag{43}$$

where

$$R_1(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(-C_5^0\zeta_3 - C_2^0\zeta_1 - 2e^{\zeta_3 d}\right)\overline{f}(\lambda)d\lambda, \tag{44}$$

$$\begin{aligned} R_2(k) = \tfrac{1}{2\pi}\textstyle\int_{-\infty}^{\infty}&\left(C_1^0\mu_1\zeta_1 + \mu_0\left(\zeta_3^2 C_5^0 + 2\zeta_3 e^{-\zeta_3 d}\right)\right. \\ &\left.+\eta\mu_0 k\left(C_5^0\zeta_3 + C_5^1\varepsilon - 2\zeta_3 e^{-\zeta_3 d}\right)\right)\overline{f}(\lambda)d\lambda. \end{aligned} \tag{45}$$

By separating Equations (34) and (41–43) into terms containing ε and those independent of ε, we obtain a system of 10 nonhomogeneous linear equations:

$$\begin{aligned} &C_1^0\cos\left(\zeta_1 H\right) - C_2^0\sin\left(\zeta_1 H\right) = 0, \\ &C_1^1\cos\left(\zeta_1 H\right) - C_2^1\sin\left(\zeta_1 H\right) = 0, \\ &C_3^0\cos\left(\zeta_2 H\right) - C_4^0\sin\left(\zeta_2 H\right) = T_0, \\ &C_3^1\cos\left(\zeta_2 H\right) - C_4^1\sin\left(\zeta_2 H\right) = 0, \\ &C_3^0 + C_4^0\zeta_2\varepsilon f(x) = T_0, C_3^1 + C_4^1\zeta_2\varepsilon f(x) = 0, \\ &C_1^0 - C_5^0 = \frac{2}{\zeta_3}e^{-\zeta_3 d}, C_1^1 - C_5^1 = R_1, \\ &\mu_1 C_2^0 + \eta\mu_2 C_5^0 = 2\mu_2 e^{-\zeta_3 d} - \frac{2}{\zeta_3}\mu_2\eta e^{-\zeta_3 d}, \end{aligned}$$

$$\mu_1 C_2^1 + \eta\mu_2 C_5^1 = R_2. \tag{46}$$

Solving the above system of equations, the values of the unknown variables are determined as follows:

$$\begin{aligned}
C_1^0 &= \frac{2\mu_0 e^{-\zeta_3 d}\tan\left(\zeta_1 H\right)}{E}, C_2^0 = \frac{2\mu_0 e^{-\zeta_3 d}}{E},\\
C_5^0 &= \frac{2e^{-\zeta_3 d}\left(\mu_0\zeta_3\tan\left(\zeta_1 H\right) - E\right)}{E\zeta_3}, C_3^0 = T_0,\\
C_4^0 &= T_0\left(\cot\zeta_2 H - \csc\zeta_2 H\right), C_3^1 = C_4^1 = 0,\\
C_1^1 &= \frac{\left(R_2 + \eta\mu_0 R_1\right)\tan\left(\zeta_1 H\right)}{E}, C_2^1 = \frac{\left(R_2 + \eta\mu_0 R_1\right)}{E},\\
C_5^1 &= \frac{\left(R_2 + \eta\mu_0 R_1\right)\tan\left(\zeta_1 H\right) - R_1 E}{E},
\end{aligned} \tag{47}$$

where

$$E = \mu_1 + \mu_0\eta\tan\zeta_1 H.$$

Substituting the values of these unknown variables in (29–31), we obtain the displacement vector for M_1 as

$$\begin{aligned}
u_2^0(x,z) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(C_1\cos\left(\zeta_1 z\right) + C_2\sin\left(\zeta_1 z\right)\right)e^{-i\eta x}d\eta\\
u_2^0 &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{1}{E}\left[2\mu_0 e^{\xi_3 d} + \varepsilon\left(R_2 + \eta\mu_0 R_1\right)\right]\right.\\
&\quad\left.\left[\tan\left(\zeta_1 H\right)\cos\left(\zeta_1 z\right) + \sin\left(\zeta_1 z\right)\right]e^{-ikx}\right)dk,\\
u_2^0 &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{2\mu_0 e^{\xi_3 d}}{E}\left[1 + \frac{\varepsilon\left(R_2 + \eta\mu_0 R_1\right)e^{\xi_3 d}}{2\mu_0}\right]\right.\\
&\quad\left.\left[\tan\left(\zeta_1 H\right)\cos\left(\zeta_1 z\right) + \sin\left(\zeta_1 z\right)\right]e^{-ikx}\right)dk.
\end{aligned} \tag{48}$$

Similarly, by putting the value of unknown variables from Equation (46), the expression for temperature distribution is obtained as

$$\begin{aligned}
T_2^0(x,z) &= \frac{1}{2\pi}\\
&\int_{-\infty}^{\infty}\frac{T_0}{\sin\left(\zeta_2 H\right)}\left(\cos\left(\zeta_2 z\right)\sin\left(\zeta_2 H\right)\right.\\
&\quad\left. + \left(\cos\left(\zeta_1 H\right) - 1\right)\sin\left(\zeta_2 z\right)\right)e^{-i\eta x}d\eta\\
T_2^0 &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{T_0}{\sin\left(\zeta_2 H\right)}\left(\sin\left(\zeta_2(z+H)\right)\right.\\
&\quad\left. - \sin\left(\zeta_2 z\right)\right)e^{-i\eta x}dk.
\end{aligned} \tag{49}$$

By applying Fourier transformation on Equation (1), $f(x)$ can be written as

$$f(\lambda) = \frac{8}{s\lambda^3}(\sin(\lambda s) - \lambda s\cos(\lambda s)). \tag{50}$$

Rewriting Equation (50) with the help of Equation (1) as

$$f(\lambda) = \frac{4ls}{\varepsilon}\left(\frac{\sin(\lambda s) - \lambda s\cos(\lambda s)}{(s\lambda)^3}\right). \tag{51}$$

Using Equations (44) and (45) along with Equations (48) and (51), $\left(R_2 + \eta\mu_0 R_1\right)$ is simplified as

$$\begin{aligned}
R_2 + \eta\mu_0 R_1 &= \frac{2sl\mu_0}{\varepsilon}\int_0^{\infty}[g(k-\lambda) + g(k+\lambda)]\\
&\quad\sqrt{\frac{\pi}{2}}\cdot\frac{1}{(\lambda s)^{3/2}}\cdot\sqrt{\frac{2}{\pi\lambda s}}\frac{\sin(\lambda s) - \lambda s\cos(\lambda s)}{(\lambda s)}d\lambda\ ,\\
&= \frac{2sl\mu_0}{\varepsilon}\int_0^{\infty}\sqrt{\frac{\pi}{2}}[g(k-\lambda) + g(k+\lambda)]\frac{J_{3/2}(\lambda s)}{(\lambda s)^{3/2}}d\lambda
\end{aligned} \tag{52}$$

where $J_{3/2}(\lambda s)$ and $g(k-\lambda)$ are given as

$$\left.\begin{aligned}
J_{3/2}(\lambda s) &= \sqrt{\tfrac{2}{\pi}}\tfrac{\sin(\lambda s) - \lambda s\cos(\lambda s)}{(\lambda s)^{3/2}}\\
g(k-l) &= \left(A_1 + A_2\right)\\
A_1 &= C_2^0\zeta_1 - C_5^0\zeta_3 - \mu_0\zeta_3\left(\zeta_3 C_5^0 - 2e^{-\zeta_3 d}\right)\\
A_2 &= -\mu_0\lambda k\left(\zeta_3 C_5^0 - 2e^{-\zeta_3 d}\right)
\end{aligned}\right\}. \tag{53}$$

Now, simplifying the integral in Equation (48) and applying the asymptotic formulas given by Willis [31] and Tranter [32], the term involving $1/s$ and its higher powers are ignored (as s is very large). Consequently, the integral is simplified as

$$\int_0^{\infty}\sqrt{\frac{\pi}{2}}\frac{1}{(\lambda s)^{3/2}}[g(k-\lambda) + g(k+\lambda)]J_{3/2}(\lambda s)d\lambda = \frac{2}{3s}g(k). \tag{54}$$

Thus, from Equation (54), Equation (52) becomes

$$R_2 + \mu_0\zeta_3 R_1 = \frac{4l\mu_0}{3\varepsilon}g(k). \tag{55}$$

From Equation (48), the displacement vector of M_1 changes to

$$u_2^0 = \frac{4\mu_0}{3\pi}\int_{-\infty}^{\infty}\left(\frac{e^{-\zeta_3 d}}{E\left[1 - lg(k)e^{\zeta_3 d}\right]}\left[\tan\left(\zeta_1 H\right)\cos\left(\zeta_1 z\right) + \sin\left(\zeta_1 z\right)\right]e^{-ikx}\right)dk. \tag{56}$$

The value of the above integral depends on the poles of the integrand, that is, putting the denominator equals to 0. Thus, the dispersion equation is given by the roots of the equation:

$$E\cdot\left[1 - lg(k)e^{\zeta_3 d}\right] = 0. \tag{57}$$

Solving Equation (58), we get

$$\tan\left(\zeta_1 H\right) = \frac{\mu_0\zeta_3 + \mu_0 l\zeta_3^2 e^{\zeta_3 d}}{l\mu_0\zeta_3 - l\mu_1\zeta_1 + \mu_1\zeta_1}. \tag{58}$$

Using Equation (26) and substituting $\omega = c_L k, \eta = k, \zeta_1 = P_1 k, q = Qk, \xi_2 = P_2 k$, and $\zeta_3 = P_3 k$, as

$$\begin{aligned}
\zeta_1^2 &= \left(\frac{\omega^2}{c_t^2} - \eta^2\right) = \left(\frac{c_L^2}{c_t^2} - 1\right)k^2, P_1 = \sqrt{\frac{c_L^2}{c_t^2} - 1},\\
\zeta_3 &= \frac{1}{2}\left(Qk + k\sqrt{Q + 4\left(1 - \frac{c_L^2}{\beta^2}\right)}\right),\\
P_3 &= \frac{1}{2}\left(Q + \sqrt{Q + 4\left(1 - \frac{c_L^2}{\beta^2}\right)}\right).
\end{aligned}$$

TABLE 1 | Numerical values of parameters.

Parameters	Numerical values	Parameters	Numerical values
μ_1	7.5×10^9 N/m^2	K	1.5×10^2 W m^{-1} K^{-1}
ρ_1	2100 kg/m^3	ρ_2	5563 kg/m^3
$c_v^{(1)}$	1.5×10^{10} J m^{-3} K^{-1}	μ_0	7.84×10^9 N/m^3
T_0	373 K	λ_2	1.3×10^{10} Pa
λ_1	1.5×10^{10} Pa	$c_v^{(2)}$	1.609×10^{10}J m^{-3} K^{-1}

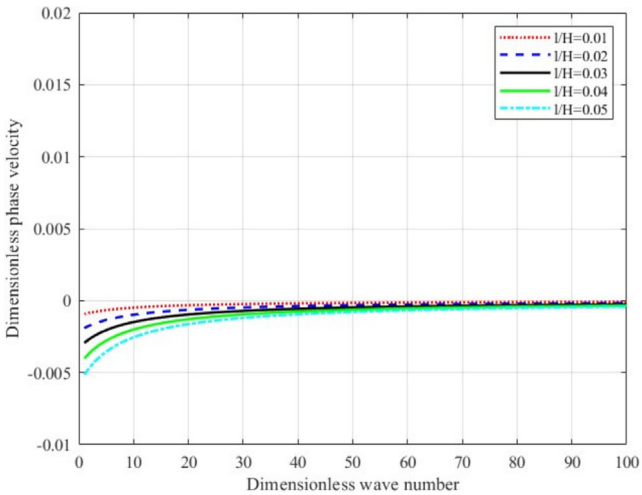

FIGURE 2 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number for the inhomogeneity parameter $q = 0$. [Colour figure can be viewed at wileyonlinelibrary.com]

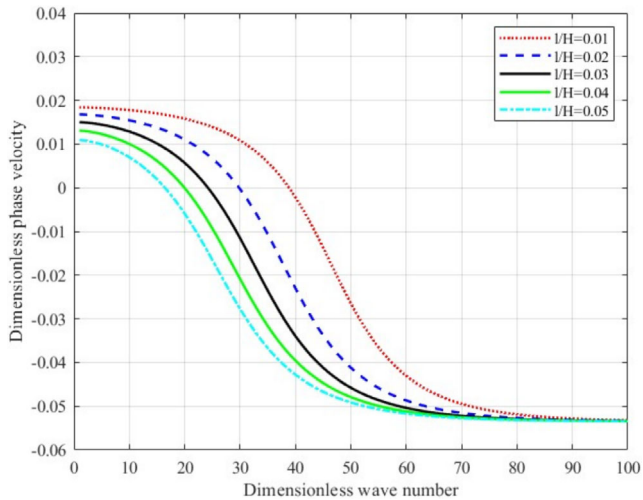

FIGURE 3 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number for the inhomogeneity parameter $q = 1$. [Colour figure can be viewed at wileyonlinelibrary.com]

Using these values, the dispersion equation for the Love waves becomes

$$\tan\left(P_1 kH\right) = \frac{\mu_0 P_3\left(1 + lP_3 k e^{P_3 kd}\right)}{\mu_1 P_1(1-l) + \mu_0 l P_3}. \tag{59}$$

This simplification makes it easier to understand how wave phase velocity relates to wave number and other parameters of the layer.

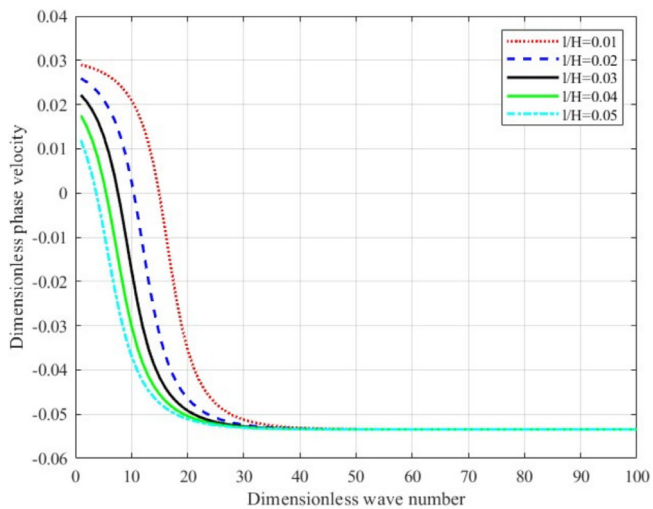

FIGURE 4 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number for the inhomogeneity parameter $q = 2$. [Colour figure can be viewed at wileyonlinelibrary.com]

7 | Numerical Results and Discussion

This section presents the numerical analysis of Love wave propagation in an isotropic thermoelastic layer overlying a nonhomogeneous elastic half-space. The dispersion relation derived in Equation (59) has been solved and graphically illustrated using MATLAB to explore the wave characteristics under various physical conditions. Material and thermal parameters for the thermoelastic layer are adopted from Kumar et al. [14], while the elastic constants for the nonhomogeneous half-space are taken from Gubbins [33]. The selected values are detailed in Table 1. These parameter choices ensure consistency with established literature and provide a realistic basis for evaluating the effects of thermoelasticity, inhomogeneity, and interfacial irregularities on phase velocity and dispersion behavior.

The influence of key parameters, such as material inhomogeneity and the ratio of irregularity depth to layer height, on the phase velocity of Love waves is investigated through Figures 2–7.

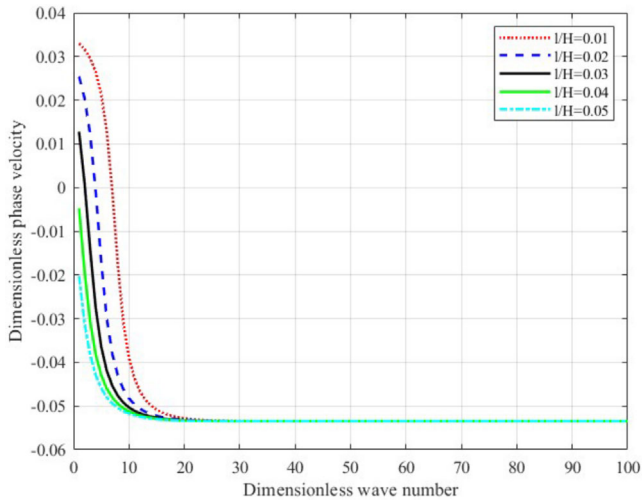

FIGURE 5 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number for the inhomogeneity parameter $q = 3$. [Colour figure can be viewed at wileyonlinelibrary.com]

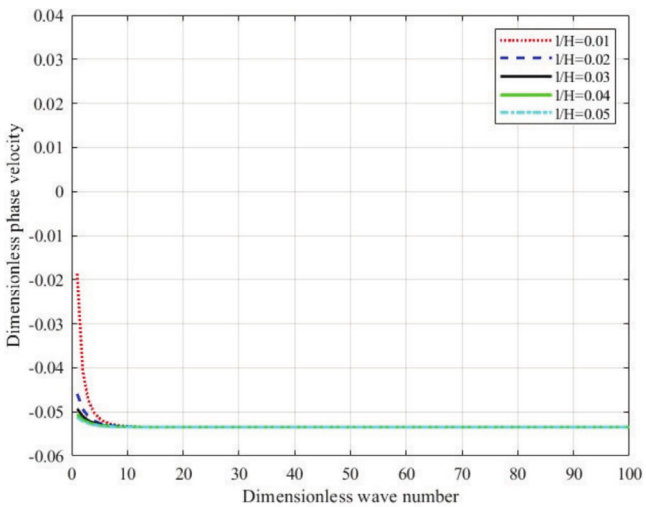

FIGURE 6 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number for the inhomogeneity parameter $q = 5$. [Colour figure can be viewed at wileyonlinelibrary.com]

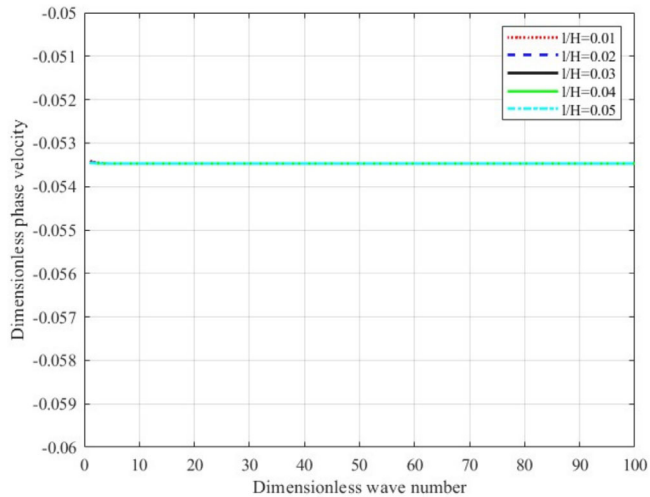

FIGURE 7 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number for the inhomogeneity parameter $q = 10$. [Colour figure can be viewed at wileyonlinelibrary.com]

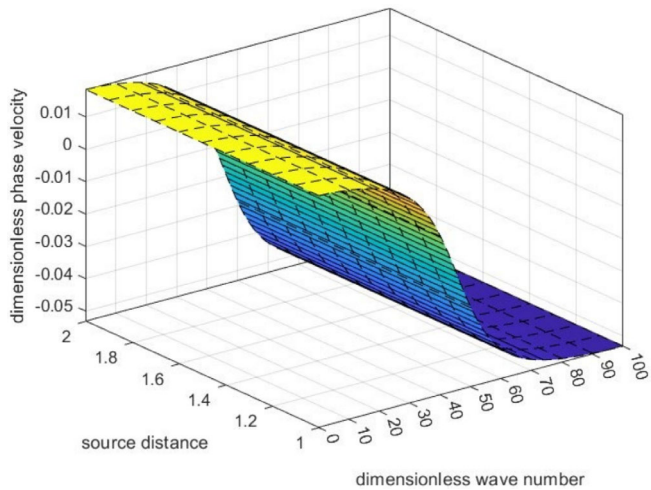

FIGURE 8 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number and point-source distance for the inhomogeneity parameter $q = 1$. [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 2 illustrates the variation of phase velocity of Love waves with respect to wave number in a thermoelastic layer overlying a homogeneous half-space, considering $q = 0$. Each curve represents a different ratio of irregularity depth to layer height, specifically $l/H = 0.01, 0.02, 0.03, 0.04$, and 0.05. A noticeable decrease in phase velocity is observed as the wave number and ratio l/H increases. This pattern indicates that increase in size of surface irregularities at the interface reduce the phase velocity, likely due to increased energy scattering and coupling effects at the irregular boundary.

Figures 3–7 present this variation for a nonhomogeneous half-space, with inhomogeneity parameters $q = 1, 2, 3, 5$, and 10, respectively. In each case, a consistent reduction in phase velocity is observed as both the wave number and irregularity ratio increase. Moreover, a notable observation is that the rate of decrease in phase velocity becomes more pronounced with higher values of q, suggesting that material inhomogeneity significantly influences wave dispersion. As q increases, the convergence of phase velocity curves for different irregularity ratios occurs at lower wave numbers, around 90, $3+5$, and 25 for $q = 1, 2, 3$, and progressively down to about 10 for $q = 5$. For $q \geq 5$, the curves nearly overlap (Figure 7), demonstrating that material inhomogeneity suppresses the impact of interfacial roughness on dispersion. This behavior highlights the sensitivity of surface wave propagation to both geometric and material irregularities in layered media.

Additionally, the results indicate that increasing the ratio of irregularity depth to the height of the thermoelastic layer leads to a reduction in phase velocity. Furthermore, inhomogeneity plays a significant role in modifying the dispersion characteristics of the Love waves.

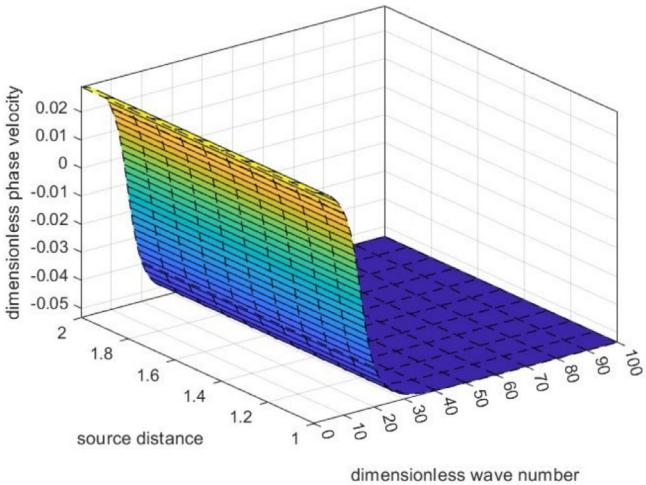

FIGURE 9 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number and point-source distance for the inhomogeneity parameter $q = 2$. [Colour figure can be viewed at wileyonlinelibrary.com]

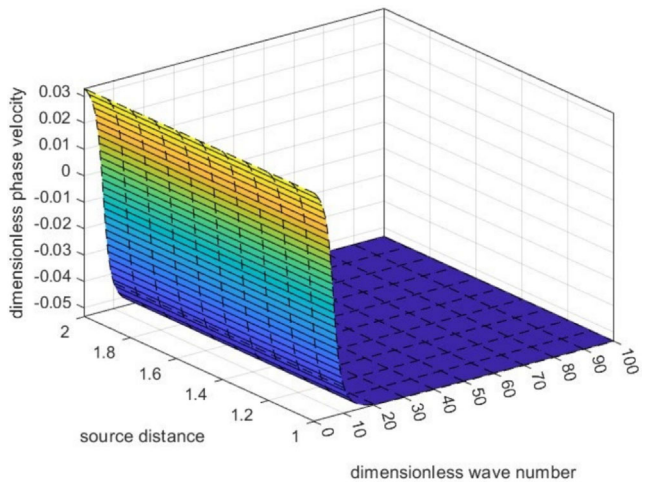

FIGURE 10 | Dispersion curves for dimensionless phase velocity versus dimensionless wave number and point-source distance for the inhomogeneity parameter $q = 3$. [Colour figure can be viewed at wileyonlinelibrary.com]

Figures 8–10 explore the influence of the point-source location on the phase velocity and wave number in a nonhomogeneous half-space for $q = 1, 2, 3$. These plots reveal that the source distance plays a vital role in the wave dynamics. For $q = 1, 2, 3$, phase velocity decays nonlinearly with source distance, with steeper declines in more inhomogeneous media.

8 | Conclusion

This study investigates Love wave propagation in a thermoelastic layer overlying a nonhomogeneous elastic half-space, featuring a parabolic interface irregularity induced by a point source located within the half-space. The dispersion equation for Love waves is derived using Fourier and inverse Fourier transforms, followed by the application of Eringen's perturbation method. By combining analytical and numerical approaches to explore thermoelastic effects, material inhomogeneity, and interfacial irregularity, the model provides a realistic framework for analyzing surface wave behavior in complex geological media. The key findings are summarized as follows:

1. The phase velocity of Love waves decreases with increasing wave number across all cases of nonhomogeneous elastic half-space, while it is increasing for $q = 0$.
2. As the inhomogeneity parameter q increases from 1 to 10, the phase velocity shows a more rapid decline with increasing wave number, with convergence points shifting from 90 ($q = 1$) to 10 ($q = 5$). This indicates a stronger dispersive effect in more inhomogeneous media.
3. Increasing the depth ratio of irregularity to the layer's height (l/H) systematically decreases phase velocity, particularly for weakly inhomogeneous media ($q \leq 2$). However, in highly graded half-spaces, material heterogeneity dominates and suppresses the influence of interface irregularity.
4. Thermal–mechanical interaction contributes to enhanced velocity reduction at shorter wavelengths through energy dissipation mechanisms.
5. The source distance also impacts wave propagation, as seen in Figures 8–10. With increasing distance of the point source from the irregular interface, the phase velocity and wave number both vary, highlighting the sensitivity of wave characteristics to source positioning in nonhomogeneous media.

This study can be extended by considering anisotropic or poroelastic materials, frequency-dependent damping, or nonparabolic interfacial shapes. Further validation through comparison with experimental or field data would also strengthen the applicability of the model to geophysical and engineering problems.

9 | Practical Application

The present model offers potential applications in geophysics, particularly in improving the interpretation of seismic data in layered and irregular geological media. By accounting for thermoelastic effects, material inhomogeneity, and interfacial irregularities, the study provides a more nuanced understanding of Love wave propagation, which can aid seismologists in modeling near-surface wave behavior more accurately. While primarily theoretical, the findings may support practical efforts in earthquake risk analysis, subsurface imaging for resource exploration, and the refinement of ground-motion models used in structural safety assessments. Additionally, the model could inform the design of more sensitive wave-based geophysical sensors and monitoring tools.

Author Contributions

Abhilasha Saini: conceptualization, methodology, writing – original draft, data curation, formal analysis, validation, investigation, soft-

ware. **Ravinder Kumar:** conceptualization, writing – review and editing, visualization, supervision, formal analysis, validation, investigation, methodology.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

No data are associated in the manuscript.

References

1. A. E. H. Love, *Some Problems of Geodynamics* (Cambridge University Press, 1911).

2. W. M. Ewing, W. S. Jardetzky, F. Press, and A. Beiser, "Elastic Waves in Layered Media," *Physics Today* 10, no. 12 (1957): 27–28.

3. K. F. Graff, *Wave Motion in Elastic Solids* (Courier Corp., 2012).

4. H. Power, "Linear Waves at the Free Surface of a Saturated Porous Media," *Zeitschrift für Angewandte Mathematik und Physik* 44, no. 3 (1993): 578–586.

5. J. Achenbach, *Wave Propagation in Elastic Solids* (Elsevier, 2012).

6. A. Chattopadhyay and B. K. Kar, "Love Waves due to a Point Source in an Isotropic Elastic Medium Under Initial Stress," *International Journal of Non-Linear Mechanics* 16, no. 3–4 (1981): 247–258.

7. M. A. Biot, "Thermoelasticity and Irreversible Thermodynamics," *Journal of Applied Physics* 27, no. 3 (1956): 240–253.

8. H. W. Lord and Y. Shulman, "A Generalized Dynamical Theory of Thermoelasticity," *Journal of the Mechanics and Physics of Solids* 15, no. 5 (1967): 299–309.

9. A. E. Green and K. Lindsay, "Thermoelasticity," *Journal of Elasticity* 2, no. 1 (1972): 1–7.

10. A. E. Green and P. Naghdi, "Thermoelasticity Without Energy Dissipation," *Journal of Elasticity* 31, no. 3 (1993): 189–208.

11. J. Bear, S. Sorek, G. Ben-Dor, and G. Mazor, "Displacement Waves in Saturated Thermoelastic Porous Media. I. Basic Equations," *Fluid Dynamics Research* 9, no. 4 (1992): 155–171.

12. R. Kumar and A. Saini, "Influence of Parabolic Irregularity, Inhomogeneity, Initial Stress, and Anisotropy on Love Wave Propagation," *Mechanics of Solids* 59, no. 3 (2024): 1443–1458.

13. A. Saini and R. Kumar, "Effect of Rigidity and Parabolic Irregularity on Love Wave Propagation in Transversely Isotropic Fluid-Saturated Porous Layer Lying Over a Nonhomogenous Half-Space," *Mechanics of Solids* 59, no. 2 (2024): 1094–1107.

14. D. Kumar, D. Singh, and S. K. Tomar, "Love-Type Waves in Thermoelastic Solid With Double Porosity Structure," *Waves in Random and Complex Media* 32, no. 1 (2022): 1–29.

15. R. Kumar, D. K. Madan, and J. S. Sikka, "Shear Wave Propagation in Multilayered Medium Including an Irregular Fluid Saturated Porous Stratum With Rigid Boundary," *Advances in Mathematical Physics* 2014 (2014): 163505.

16. R. Kumar, D. K. Madan, and J. S. Sikka, "Wave Propagation in an Irregular Fluid Saturated Porous Anisotropic Layer Sandwiched Between a Homogeneous Layer and Half-Space," *WSEAS Transactions on Applied and Theoretical Mechanics* 10 (2015): 62–70.

17. A. Saini and R. Kumar, "Analysis of Love Waves in an Initially Stressed Transversely Isotropic Porous Layer Over Heterogeneous Half Space With Parabolic Irregularity," *Journal of Vibration Engineering & Technology* 12 (2024): 7373.

18. R. Kumar and A. Saini, "Effect of Anisotropy, Inhomogeneity and Porosity on Love Wave Propagation Through Fluid-Saturated Porous Layers in Irregular Layered Media," *European Physical Journal Plus* 139, no. 12 (2024): 1–20.

19. J. Lalvohbika and S. S. Singh, "Scattering of qSH-Waves From a Corrugated Interface Between Two Dissimilar Nematic Elastomers," *Waves in Random and Complex Media* 32, no. 1 (2022): 19–42.

20. S. S. Singh, "Love Wave at a Layer Medium Bounded by Irregular Boundary Surfaces," *Journal of Vibration and Control* 17, no. 5 (2011): 789–795.

21. S. S. Singh and Z. Oinam, "Haskell Matrix Method for the Propagation of Love Waves in Multilayers of Magneto-Thermoelastic Materials," *International Journal of Applied Mechanics* 17, no. 2 (2025): 2550004.

22. R. Kumar, S. Sharma, and S. Chandel, "Impact of Triangular Irregularity, Material Heterogeneity and Initial Stress on the Propagation of Shear Waves in a Transversely Isotropic Porous Layer," *International Journal of Applied Mechanics and Engineering* 30, no. 2 (2025): 89–104.

23. H. Sharma, S. Kumari, and P. Ailawalia, "Wave Propagation in Thermo-Viscoelastic Semiconducting Medium With Hydrostatic Initial Stress," *Mechanics of Solids* 36 (2024): 1553–1564.

24. S. Kumar, H. Nagar, and S. Kumari, "Reflection of Plane Wave at the Interface Between Thermoelastic and Hygro-Thermoelastic Solid Half-Space," *Mechanics of Solids* 60, no. 1 (2025): 384–402.

25. K. Gajroiya and J. S. Sikka, "Reflection and Transmission Characteristics of Plane-P-Wave at the Interface of Anisotropic Nonlocal Poroelastic and Thermoelastic Solid Half-Spaces," *Acta Mechanica* 235 (2024): 1–26.

26. R. Kumar, A. Goyal, and A. Saini, "Effect of Inhomogeneity and Triangular Irregularity on Propagation of Shear Waves in an Anisotropic Porous Layer," *Mechanics of Solids* 59, no. 2 (2024): 1036–1049.

27. K. Gajroiya, N. Malik, and J. S. Sikka, "Deformation of a Rotating Porothermoelastic Medium With an Internal Moving Heat Source Under Thermal Shock and Mechanical Load," *Journal of Vibration Engineering & Technologies* 13, no. 1 (2025): 133–148.

28. P. Ailawalia, S. K. Sachdeva, and D. S. Pathania, "Plane Strain Deformation in a Thermoelastic Microelongated Solid With Internal Heat Source," *International Journal of Applied Mechanics and Engineering* 20, no. 4 (2015): 717–731.

29. R. K. Sahrawat and K. Kumar, "Plane Wave and Fundamental Solution in Non-Local Couple Stress Micropolar Thermoelastic Solid Without Energy Dissipation," *Journal of Thermal Stresses* 44 (2020): 295–314.

30. A. C. Eringen and J. C. Samuels, "Impact and Moving Loads on a Slightly Curved Elastic Half-Space," *Journal of Applied Mechanics* 26, no. 4 (1959): 491–498.

31. H. F. Willis, "A Formula for Expanding an Integral as a Series," *Philosophical Magazine* 39, no. 293 (1948): 455–459.

32. C. J. Tranter, *Integral Transforms in Mathematical Physics* (Methuen & Co., 1966), 63–67.

33. D. Gubbins, *Seismology and Plate Tectonics* (Cambridge University Press, 1990).